

AI Voice Studio

A Privacy-First Voice Intelligence Platform
for the Health Sector

Product Concept Briefing

May 2025

Classification:

Internal — Confidential

Audience:

Internal Team

OVERVIEW

Executive Summary

This briefing outlines a concept for a proprietary AI-powered voice content platform designed specifically for the health sector. The platform enables organisations to create professional-grade audio content — such as clinical briefings, patient education material, and staff communications — using personalised AI voice personas trained on real human voices.

The core differentiator is a **privacy-first architecture**: all voice data, transcripts, and generated audio remain under the organisation's direct control, satisfying the requirements of the NZ Privacy Act 2020, the Health Information Privacy Code 2020, and Health New Zealand procurement standards.

***The opportunity:** Health organisations produce significant volumes of written content (clinical guidelines, patient communications, training material) that could reach staff and patients more effectively as on-demand audio. No compliant, affordable, local solution exists for this today.*

THE PROBLEM

Health Sector Content Has a Last-Mile Problem

Clinical teams produce enormous volumes of written content — guidelines, procedure updates, patient education, training material — that frequently goes unread. Audio is more accessible, more engaging, and more inclusive. Yet health organisations face a critical blocker:

Data sovereignty

Health information cannot be processed by external cloud services without a documented risk assessment and patient/staff consent. Most commercial AI voice tools send data offshore.

Voice quality gap

Free or open-source tools produce robotic, unconvincing audio that undermines professional credibility.

No personalisation

Generic AI voices lack the authority and familiarity of a known clinician or trusted organisation voice.

Compliance overhead

Building a compliant in-house solution has historically required specialist AI engineering expertise that health organisations do not have.

SOLUTION

What We Are Building

A self-contained, web-based platform that allows health organisations to:

1 Train a voice persona

A clinician, presenter, or brand voice records a short audio sample. The platform learns that voice and stores it securely within the organisation's own infrastructure.

2 Generate audio from any transcript

Staff paste or upload a written document. The platform rewrites it into natural, conversational spoken language and synthesises it in the chosen voice.

3 Choose the format

Single narrator for briefings and education material, or a two-voice conversation format for more engaging content — similar to a professional podcast.

4 Manage voice libraries

Organisations build a library of approved voices — both individual staff personas and official organisational voices — that can be used across the team.

QUALITY

Premium Audio — Not Robotic

The quality gap between consumer AI voice tools and professional broadcast audio has historically been a barrier to adoption. This platform closes that gap through a layered approach:

- Natural language rewriting — the system transforms written text into spoken language before synthesis, adding natural pacing, conversational rhythm, and appropriate emphasis
- Emotional intelligence — the audio engine interprets context to vary tone, warmth, and energy across the content rather than reading at a flat, uniform pace
- Broadcast-standard audio processing — the final output is professionally processed to match the loudness and clarity standards used in commercial radio and podcasting
- Voice consistency — trained personas sound consistent across any content, reinforcing organisational voice and brand

PRIVACY & GOVERNANCE

Built for NZ Health Compliance

Privacy is not an add-on — it is the architectural foundation. The platform is designed to meet the specific requirements of the NZ regulatory environment:

Requirement	How it is addressed
NZ Privacy Act 2020	All personal data processed and stored within the organisation's own infrastructure
Health Information Privacy Code 2020	Voice samples treated as sensitive biometric data — encrypted, access-logged, and deletable on request
Health NZ procurement standards	Two deployment options available to match each organisation's risk appetite and infrastructure
Staff consent	Built-in consent workflow — staff explicitly consent before any voice persona is created, with a logged timestamp
Audit trail	Every action (who created what, when, using which voice) is logged in an immutable audit record
Access control	Role-based permissions — org administrators, content creators, and viewers each have appropriate access levels
Data breach readiness	Monitoring and alerting aligned with the 72-hour Privacy Commissioner notification requirement

DEPLOYMENT

Two Options — One Decision

The platform is designed to work in either of two deployment modes, depending on the organisation's infrastructure and risk appetite. The same product, the same interface, the same quality — the difference is where the computation happens.

	Option A Fully On-Premises	Option B Managed Cloud (NZ-Hosted)
Where data lives	Inside the organisation's own server room — nothing leaves the building	Within a New Zealand-based data centre, under a formal data processing agreement
Internet required	No — fully air-gapped capable	Yes — for content generation calls
Infrastructure	On-premise GPU server (organisation-owned)	Thin on-premise layer; compute handled by NZ-hosted cloud
Compliance path	Straightforward — no third-party data sharing occurs	Requires a Privacy Impact Assessment and data processing agreement prior to go-live
Voice quality	Very high — approaching commercial studio quality	Commercial studio quality — equivalent to leading market products
Best suited for	Organisations with a strict no-external-cloud policy	Organisations already using managed cloud services and Microsoft 365

Note: The architecture is designed so that both deployment options are supported from a single codebase. An organisation can start with Option B and migrate to Option A as their infrastructure matures, without rebuilding the product.

OPPORTUNITY

Why This, Why Now

The global AI voice market is growing rapidly, but the health sector remains significantly underserved — specifically because existing tools cannot meet its data sovereignty requirements. This creates a clear window:

- **No compliant local alternative exists.** Commercial AI voice platforms (including the market leader) operate on external cloud infrastructure incompatible with strict health data requirements.
- **Demand is real and increasing.** Health organisations are actively seeking ways to make content more accessible to staff and patients. Audio is a proven format for engagement and retention.

- **The technology is now mature enough.** Open, production-grade AI voice technology has reached a quality threshold where the gap between local and commercial cloud solutions is negligible.
- **Microsoft partnership creates a deployment advantage.** For organisations already in the Microsoft ecosystem, we can offer a compliant, integrated path to deployment using existing trusted infrastructure.
- **First-mover advantage is significant.** Health sector software procurement cycles are long. Being the established compliant solution creates durable competitive positioning.

SUCCESS CRITERIA

What Good Looks Like at Launch

A successful MVP delivers the following demonstrable outcomes:

Outcome	Measure
Voice persona quality	A listener cannot reliably distinguish the AI voice from the source recording in a blind test
Generation speed	A 5-minute audio piece is generated in under 2 minutes from transcript upload
Data sovereignty	Zero external API calls during content generation — verified with network monitoring
Compliance readiness	Product ships with a completed Data Flow Map and Privacy Impact Assessment template for client use
Usability	A non-technical staff member can create a voice persona and generate their first audio piece within 15 minutes, without training

NEXT STEPS

Proposed Path Forward

- 1

Alignment

Team review of this briefing. Agree on the scope of the MVP and which deployment option to prioritise for the initial pitch.
- 2

Proof of Concept

Internal build of the core voice cloning and audio generation pipeline. Goal: produce a demonstration using a real team member's voice on a sample health transcript.
- 3

Client Identification

Identify one or two anchor health sector organisations to engage as design partners. Their requirements will shape the compliance documentation and governance workflow.
- 4

Privacy & Legal Review

Engage legal/privacy counsel to confirm the platform design against NZ HIPC requirements before any client data is processed.

5

Pilot Deployment

Deploy to one design partner organisation in a controlled environment. Gather feedback on voice quality, usability, and workflow fit.

6

Commercial Packaging

Define pricing model, SLA structure, and support offering before broader market engagement.

This document is intended for internal discussion only. The product concept, architecture approach, and commercial strategy described herein are proprietary and confidential. Please do not share outside the team.

Questions or feedback? Reach out to the project lead directly.